

Whiteout (Arctic / Snow / Visibility, Thermal Sensors, Navigation)

Executive Summary

Whiteout is a snow-themed Arctic battlefield tool for mid-to-large-scale combat, the thermal sensor warfare, and hazard avoidance. The terrain features blizzard corridors, ice dunes, frozen ravines, research shelters, buried comms towers, and wind-eroded ridges. The map emphasizes Obsidian Protocol's autonomous navigation logic under near-zero visibility, thermal sensor fusion, comms-radar warfare, and tactics under severe Arctic conditions. In AR (Realistic/Lite/Lite), the map anchors at ~4-5 m viewrange distance, cover a 10x10 m physical area representing ~1 km² of Arctic terrain. Key features include whiteout zones, thermal blind pockets, unstable ice routes, and navigation-driven tactical decision-making.

Design Goals & Gameplay

Visibility-Corrupted Warfare

Whiteout conditions combat in near-zero visibility. Blizzards, drifting snow, and ice fog reduce sightlines to a few meters. Units rely heavily on thermal sensors and autonomous pathfinding.

Thermal-Sensor Dominance

Heat signatures become the primary detection method. Terrain features like ice ridges, snow berms, and geothermal vents distort thermal readings, creating deceptive pockets.

Autonomy & Hazard Navigation

Autonomous units choose routes based on wind exposure, ice stability, thermal cover, and sensor clarity. Units avoid whiteout corridors unless strategically necessary.

Sensor & Information Warfare

Blizzards, ice fog, thermal distortions, and magnetic interference degrade sensors. Relay towers on ridges restore clarity. Destroying relays creates massive blind pockets across the snowfields.

Logistics & Resources

Fuel caches, heated shelters, and research stations serve as resupply nodes. Convoys must navigate unstable ice and avoid ambush-prone snow corridors.

Deployment & Progress

A hard-deployment budget (~10,000 points) forces players to choose between thermal-optimal infantry, Arctic drones, or heavy snow-eligible armor. Blue deploys from the southwest snowfield camp; Red deploys from the northeast ridge shelter.

Spatial Constraints (Nreal AR)

Device Capabilities

Target resolution: ~10° diagonal FOV, 1020x1080 per eye resolution. Optimal viewing distance: ~4-6 m.

Playable Area & Scaling

10x10 m physical area = 1 km² Arctic zone (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Inside-out tracking affects horizontal planes. Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 Unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ Snowfield Camp — (0,0). 150x150 m Snow berms, heated tents, drone pods. Arctic stable.

Red HQ Ridge Shelter Command — (1000,1000). 100x100 m Heated bunker with radar mast and thermal relays.

Central Whiteout Basin — (500,500). 300x300 m Blizzard corridor: near-zero visibility, unstable ice, thermal distortions.

Research Shanty Cluster — (250,250). 120x120 m Low, open-air. Operations, thermal shelters, storage vaults.

Ice Rider Network — (800,300). 200x200 m Wind-carved ridges, snow tunnels, flak-like routes.

Blizzard Corridor — (0,500) to (1000,500) Heavy cold-air zone, extreme sensor degradation.

Frozen Trail Route — (300,2) to (450,350) Safe but slow: ice cracks and wind exposure.

Thermal Relay A — (100,100). 400 m Low-visibility sensor station.

Thermal Relay B — (150,850). 400 m Comms relay hub and weathered control node.

Heated Forward Outpost — (500,500) Snow-covered zone.

Chokepoints & Sightlines

Blizzard Corridor

Severe visibility loss, ideal for stealth ambushes and thermal deception.

Ice Ridge Networks

Unstable footing routes, unpredictable cover due to drifting snow.

Whiteout Basin

Unstable terrain, long-range thermal engagements with environmental hazards.

Frozen Trail Route

Predictable but safe, vulnerable to overwatch from ridges.

Sightlines

Thermal Relay A provides long-range sensor coverage (~200 m). Blizzards and ice fog create fog-of-war pockets.

Deployment Zones

Blue Deployment: SW snowfield camps + Arctic stable.

Red Deployment: NE ridge shelter + thermal stations.

Neutral Forward Zone: Cluster of pods and fuel drops.

Assets & Environment

Terrain

Ice ridges, snow dunes, frozen lakes, blizzard corridors, research shelters.

Vegetation

None — Arctic tundra only.

Buildings & Props

Military: Radar dishes, thermal relays, bunkers, industrial generators, pipelines, research shelters, Logistics: Trucks, crates, fuel drums, drones.

Effects

Snowstorms, ice fog, wind gusts, thermal distortion.

Audio

Wind, cracking ice, distant machinery, radio chatter.

LOD & Budget

2048 for terrain/buildings, 1024 for props, 1024 for compression. Target 4300 MB.

Texture Budget

Asset Cate	Count	Tx/Item	Total Size
Ice Texture	1	200K	200K
Shelters/Props	4	8K	32K
Buildings	15	15K	225K
Roads/Fuel	4	15K	60K
Props	50	15K	750K
Total	—	—	~1.2MB

Unreal / Nreal Integration

Engine & Plugins

Unreal 5.4 Template + Nreal SDK (NREAL), ARMR/MRCare enabled.

Rendering

Forward renderer, stationary lights, baked GI, aggressive culling.

Inputs

Inside-out anchor for AR/mix.

Interaction

Phone gesture for pass/covers, HUD as screen-space widgets.

Multitask

Client-server architecture offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype build — Month 1: Block-out whiteout basins, ridge networks, research shelters, AR anchor test.

2. Core Systems — Month 2: Hack/fix, cold-weather autonomy behaviors, thermal logic.

3. AR Integration — Month 3: Anchor, gaze-direction, basic interactivity.

4. Content Fill — Month 4-5: Final models, textures, lighting, audio.

5. Gameplay & Polish — Month 6-7: Optimization, balance, VR/AR connectivity, ice fog, thermal distortion.

6. Testing & QA — Month 8: Multiplayer stress test, sensor validation.

Testing Plan

Autonomy

Verify cold-weather navigation, thermal-sensor logic, hazard avoidance.

Comms/Sensors

Test comms/deception, fog-of-war updates, LOS checks under blizzard conditions.

Performance

100 FPS AR mode on target device.

QA Checklist

Navigation, thermal logic, deployment budget enforcement, anchor drift, sensor accuracy.